Fast Hyperspectral Image Denoising and Destriping Method Based on Graph Laplacian Regularization

Xin Su, Zhi Zhang, and Fang Yang

Abstract— **Hyperspectral images (HSIs) contain rich spatial and spectral information about the earth, and are widely used in the remote sensing field. However, an HSI is frequently corrupted by various types of noise, such as Gaussian noise, sparse noise, stripe noise, and so on, which severely limits the subsequent application of the HSI. In this article, we propose a graph Laplacian regularizer (GLR) to exploit the low-rank information across the bands of the HSI. Compared with the traditional low-rank regularization, our graph Laplacian regularization can achieve equivalent or better performance with less time consumption. Besides, a sparse constraint and a low-rank constraint are employed to remove the sparse and stripe noise. In addition, the augmented Lagrangian multiplier is used to solve each component to restore a clean image. Finally, we have carried out experiments on the simulated and real noisy HSI data. The results show the superiority of the proposed method over state-of-the-art methods, in terms of PSNR and SSIM, time cost, and visual effect. The MATLAB code is available through: https://github.com/zzhang-99/FGLR.**

Index Terms— **Graph Laplacian regularizer (GLR), graph signal processing (GSP), hyperspectral image (HSI) denoising, low-rank.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain detailed spatial and spectral information about the observed objects. Because of their broad spectral dimension, HSIs are widely applied in vast areas, such as military surveillance [1], [2], smart agriculture [3], [4], environmental monitoring [5], [6], ecological protection [7], and so on. However, during the acquisition of HSIs, different types of noises and artifacts are inevitably introduced due to the defects of remote sensing satellite sensors, the unstable working environment, etc. These noises and artifacts degrade the image quality and impede the development of subsequent applications. Therefore, to provide more reliable and cleaner data for further analysis and applications, an effective mixed noise removal approach is required.

The simplest idea to denoise an HSI is to regard each band as a 2-D image, in this way, a lot of classical natural image denoising methods can be used directly, such as the bilateral filter [8], the nonlocal-means (NLMs) algorithm [9], the block-matching 3-D filtering (BM3D) [10], etc. Another intuitive way is to view an HSI as a 3-D data cube, thus the volumetric data denoising methods can be applied, for example, BM4D [11], VBM4D [12], and PCA-BM4D [13], etc. However, these methods do not fully exploit or even ignore the spectral correlation of the HSIs, resulting in unsatisfying denoising performance, so they are not suitable for HSI denoising.

In the past decades, numerous denoising algorithms for HSIs have emerged, which can be roughly classified into three groups. The first type combines the spatial and spectral information of the HSIs directly. For example, Aggarwal and Majumdar [14] proposed to apply the total variation (TV) model to the spatial and spectral domains of HSI, respectively, i.e., using 2-D TV in the spatial domain and 1-D TV in the spectral dimension, thus removing Gaussian and sparse noise. Yuan et al. [15] considered that the noise distribution of different bands of an HSI is inconsistent and proposed a spectral-spatial adaptive TV (SSTV) model, which can be adaptively adjusted to the noise intensity in the different bands. Except for the TV model, sparse representation can be also used to combine the spatial and spectral information of an HSI. For instance, Qian and Ye [16] assumed that the noise-free component in a noisy signal can be decomposed sparsely based on a redundant dictionary. Hence, they proposed to embed a nonlocal similarity and spectral-spatial structure of the HSI into a sparse representation. Lu et al. [17] made full use of correlated spectral information and similar spatial patches to improve the estimation quality of HSIs through sparse representation. Besides, some transform domain-based methods have also been proposed [18], [19]. However, these methods mainly focus on Gaussian noise or mixed Poisson-Gaussian noise removal, and do not perform well in mixed noise removal.

The second type develops algorithms that exploit the low-rank property of the HSIs. Zhang et al. [20] regarded the noiseless HSI data as a low-rank matrix and proposed a low-rank matrix recovery (LRMR) method. LRMR does not make full use of spatial information. As a result, it loses a lot of image structure information and performs poorly in complex noise cases. To solve the deficiencies of LRMR, researchers have made a series of improvements. For example, He et al. [21] explored the spatial piecewise smooth structure of the HSI with TV regularization, and proposed

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a TV-based low-rank matrix decomposition. Zhuang and Bioucas-Dias [22] proposed an HSI denoising algorithm that can quickly remove Gaussian and sparse noise by using the extremely compact and sparse representation associated with the low-rank regularization and the self-similar features. He et al. [23] proposed a spatial-spectral TV regularized local LRMR (LLRSSTV). Chen et al. [24] came up with a group sparsity regularization of the spatial difference images for HSI restoration [group sparsity-regularized low-rank tensor decomposition (LRTDGS)]. Zhuang and Ng proposed an ℓ_1-norm based subspace representation [ℓ_1-norm-regularized HSI Mixed noise Denoiser (L1HyMixDe)] [25]. Their following work [26] introduced robust hyperspectral denoiser (RhyDe), which implemented explicit low-rank representation and retained rare pixels by using a cooperative sparse form. Furthermore, to better remove the stripe noise, both [27] and [28] considered the stripe noise as an independent component and added an additional low-rank constraint on the stripe component. Yang et al. [27] proposed a locally spectral and stripe low-rank SSTV method (SSLR-SSTV) and [28] came up with a globally spectral and stripe low-rank method, termed as DLR. The low-rank-based techniques uncover the relation between bands and yield satisfactory results. They are currently the mainstream methods for HSI denoising.

More recently, deep learning-based methods are favored in HSI denoising. These methods are mainly based on the convolution neural networks (CNNs), for example, the HSI denoising method based on residual CNN (HSID-CNN) [29], the HSI single-denoising CNN (HSI-SDeCNN) [30], the low-rank net (LR-Net) [31], etc. Although deep learning methods make a certain breakthrough in HSI noise reduction due to their powerful data representation ability, they still face many problems. In particular, extensive clean/noisy HSI pairs are required to train the model, but in reality, it is difficult to obtain a large number of high-quality data. In addition, training a model always takes a lot of time. Moreover, the training process is based on a certain noise prior, i.e., once the noise condition changes, the model cannot adapt to the new noise situation and has to be retrained from scratch. To avoid the training data issue of deep learning methods, recent research focuses on the unsupervised or self-supervised denoising method [32]. However, pretrained parameters by supervised networks are still needed for those self-supervised methods.

Graph signal processing (GSP) is an overwhelming branch in the field of signal processing [33] and has been widely used in various applications of high-dimensional data, providing a new perspective to solve signal processing problems [34]. HSIs are high-dimensional data with typically structural information and can naturally be represented by graphs. In fact, GSP has already been put into use in HSI denoising. Lu et al. [35] combined the low-rank matrix representation with a graph Laplacian regularization to denoise and destripe the HSIs. Cai et al. [36] used GSP to characterize the piecewise smoothing (PWS) properties of HSIs, and combined it with the LRMR model, thus proposing an adaptive weighted graph TV regularization algorithm [low-rank adaptive weight graph TV (LRAWGTV)]. Takemoto et al. [37] improved the SSTV model by introducing a graph that reflects the spatial structure of the HSI. Based on this graph, they proposed a new TV-type regularization called Graph-SSTV (GSSTV) for better denoising. Zhang and Yang [38] proposed a mixed noise removal method by combining sparse representation theory with a graph Laplacian regularizer (SRGLR). Despite the fact that GSP helps improve the denoising performance, there still exist some limitations. Firstly, current HSI denoising methods with GSP have high space and time complexity because their graphs have numerous vertices. Secondly, these graphs cannot fully characterize the low-rank property of an HSI, so their corresponding regularization does not contribute much in reality. For instance, in [35] and [36], the low-rank term plays a major role in reducing noise, and in [38], the data fidelity term and the constraints on various noises are the key parts. In other words, the performance of these methods will not be affected too much even without the graph regularization terms. Therefore, it still remains a big challenge to take full advantage of GSP for HSI denoising.

In this article, we try to solve the mixed noise removal issue by proposing a fast, low computation cost, and efficient HSI denoising method. To this purpose, a graph Laplacian regularizer (GLR) is used to exploit the low-rank property across the bands of the HSI. Compared with the current low-rank regularization achieved by nuclear norm, our graph Laplacian regularization is more efficient and can achieve better or equivalent performance. In addition, the proposed fast GLR (FGLR) denoising method yields better or comparable noise removal performance, with much less computational cost than existing model-driven-based methods. Compared with the deep learning-based methods, FGLR requires fewer computation operations and produces better or comparable denoising performance. Moreover, FGLR is easy to deploy to the satellites on the on-orbit terminal side for real-time operation. The main contributions of this article can be summarized as follows.

1) Unlike the usual graphs that are constructed on each pixel or patch of HSI, our graph is built based on the bands of HSI, which has a much smaller number of graph vertices. Consequently, the proposed method is much less computationally demanding.
2) Compared with the traditional low-rank regularization modeled by the nuclear norm, we propose a graph of Laplacian regularization based on the above graph. The theoretical deduction and toy examples have proven that the proposed graph Laplacian regularization is as effective as the traditional low-rank regularization models. Moreover, FGLR has a closed-form solution for recovering the clean HSI, which is more efficient that the traditional low-rank regularization achieved by nuclear norm.
3) To verify the effectiveness of the proposed method, we carry out extensive comparison experiments on both simulated HSI data and real HSI data. The visual performance and quantitative indexes, e.g., PSNR and SSIM, show that our method outperforms the state-of-the-art methods. In addition, the recovered spectral curves and mean vertical profile by our method have less

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distortion than the other methods. Furthermore, an image classification experiment is set up to testify effectiveness and usefulness of FGLR on subsequent application.

The rest of this article is arranged as follows. Section II introduces some preliminary knowledge on GSP and low-rank property of stripe noise. Section III presents our proposed optimization model based on GLR and the corresponding optimization procedure. The simulation experiment and real experiment are carried out, and the experimental results are analyzed and explained in detail in Section IV. Section V concludes this article.

II. Backgrounds and Preliminaries

A. Fundamentals of GSP

GSP is a branch of signal processing and a powerful tool for signal analysis. A graph $\mathcal{G}(\mathcal{V}, \mathcal{E}, \mathbf{W})$ consists of a set of K vertices $\mathcal{V} = \{v_k\}_{k=1}^K$, an edge set $\mathcal{E}$, and an adjacency matrix $\mathbf{W} \in \mathbb{R}^{K\times K}$ [33]. The edge connecting two vertices is weighted by $W_{i,j} \in \mathbf{W}$. Generally, the weight $W_{i,j}$ is computed via

$$W_{i,j} = \begin{cases} \exp\left(-\frac{|\phi(v_i,v_j)|^2}{2\varepsilon^2}\right), & \text{if } |\phi(v_i, v_j)| \le \xi \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $\phi(v_i, v_j)$ is a metric function that measures the distance between vertices v_i and v_j, and ξ is a given threshold. Note that $\phi(v_i, v_j)$ does not necessarily mean the Euclidean distance between two vertices.

The degree matrix $\mathbf{D}$ is a diagonal matrix, and its ith diagonal element $D_{i,i}$ is equal to the sum of the weights of all the edges that are adjacent to vertex i, i.e., $D_{i,i} = \sum_{j=1}^N W_{i,j}$. Accordingly, the graph Laplacian is defined as

$$\mathbf{L} = \mathbf{D} - \mathbf{W} \tag{2}$$

which is symmetric, positive, and semi-definite.

The graph signal $\mathbf{F} = \{\mathbf{f}_k\}_{k=1}^K$ is located on the vertices $\mathcal{V}$, and $\mathbf{f}_k$ is an n-dimensional vector, i.e., $\mathbf{f}_k \in \mathbb{R}^n$. Therefore, the graph signal can be represented by a matrix $\mathbf{F} \in \mathbb{R}^{n\times K}$.

B. Graph Laplacian Regularizer

The graph Laplacian regularizer is used to constrain the smoothness of a graph signal $\mathbf{F}$, and it can be expressed in the following quadratic form [39]:

$$S_{\mathcal{G}}(\mathbf{F}) = \frac{1}{2}\sum_{i,j=1}^K W_{i,j}\|\mathbf{f}_i - \mathbf{f}_j\|_2^2 = \mathrm{Tr}\left(\mathbf{F}L\mathbf{F}^T\right) \tag{3}$$

where $\mathrm{Tr}(\cdot)$ means the trace operation.

C. Low-Rank Regularizer of Stripe Noise

The stripe noise has strong structural and directional characteristics. To appropriately model the stripes, we analyze the stripe noise quantitatively by performing singular value decomposition (SVD) on a simulated stripe noise image. As shown in Fig. 1, the singular values drop dramatically to 0, with the rank index equal to 1, which demonstrates that the stripe noise has a low-rank property. Therefore, the stripe noise $\mathbf{S}$ of an HSI can be well modeled by the following low-rank regularization:

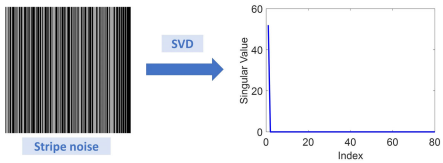


Fig. 1. Low-rank property of the stripe component.

$$R(\mathbf{S}) = \sum_{p=1}^P \mathrm{rank}\left(\mathbf{S}_p\right)$$
$$\text{s.t.} \quad \mathrm{rank}\left(\mathbf{S}_p\right) \le r \tag{4}$$

where $\mathbf{S}_p$ is the stripe noise at band p, and r is a constant, which means the upper bound of the rank of $\mathbf{S}_p$.

III. Proposed Method

A. HSI Degradation Model

A mixture of three different types of noises are considered in this article, including Gaussian noise, stripe noise, and sparse noise. Then our HSI observation model is described as follows:

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} + \mathbf{B} + \mathbf{S} \tag{5}$$

where $\mathbf{Y} \in \mathbb{R}^{M\times N\times P}$ is the observed HSI with M rows, N columns, and P band. $\mathbf{X}$ is the noise-free image, $\mathbf{N}$ represents the Gaussian noise, $\mathbf{B}$ denotes the sparse noise, and $\mathbf{S}$ indicates the stripe noise. Given the observed noise data $\mathbf{Y}$, we try to restore $\mathbf{X}$ from $\mathbf{Y}$.

B. HSI Graph Construction

As shown in Fig. 2, to construct the graph, our method takes each band of the HSI as a vertex. To fully exploit the correlation between HSI bands, we compute the weights on the edges that connect the vertices of the graph (i.e., HSI bands) by the difference of band indexes. Therefore, the weights are expressed as follows:

$$w_{i,j} = \begin{cases} \exp\left(-\frac{\|p_i - p_j\|^2}{2\varepsilon^2}\right), & \text{if } \|p_i - p_j\|^2 \le 36 \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where ε stands for the deviation of the Gaussian kernel function, p_i and p_j correspond to the index of the different HSI bands. In this way, a pair of bands with close indexes would have a high weight on the corresponding graph edge, which follows the assumption that the neighbor bands may be similar to each other [20].

C. Optimization Model

To take advantage of the correlation between the HSI bands, as well as the sparsity of the sparse noise and the low-rank properties of the stripe noise, three regularizers are combined to form the optimization model. They are the GLR $S_{\mathcal{G}}(\mathbf{F})$ defined in Section II-B, the low-rank term $R(\mathbf{S})$ defined

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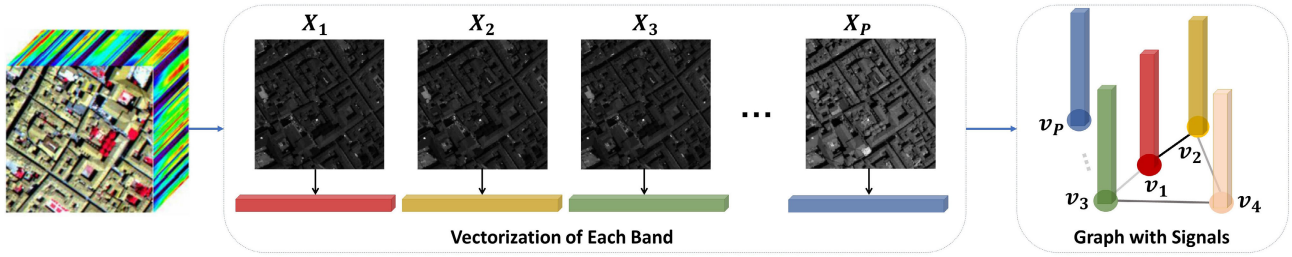


Fig. 2. Graph construction.

in Section II-C, and an $\ell_1(\cdot)$ norm to constrain the sparse noise. Thus, the following optimization model is proposed:

$$\begin{aligned} &\arg\min_{\mathbf{X},\mathbf{B},\mathbf{S}} S_{\mathcal{G}}(\mathbf{X}) + \lambda\|\mathbf{B}\|_1 + \beta R(\mathbf{S}) \\ &\text{s.t.} \quad \|\mathbf{Y}-\mathbf{X}-\mathbf{B}-\mathbf{S}\|_F^2 \le \zeta, \left\|\mathbf{Y}_p-\mathbf{X}_p-\mathbf{B}_p-\mathbf{S}_p\right\|_F^2 \le \zeta \\ &\qquad \operatorname{rank}(\mathbf{S}_p) \le r \end{aligned} \tag{7}$$

where λ and β are the coefficients for constraining the sparse and the stripe noise. ζ is a small value of the fidelity term, and r is the low-rank upper limit of the stripe noise.

Since the low-rank constraint rank$(\cdot)$ in the above formula is nonconvex, the nuclear norm $\|\cdot\|_*$ is usually used as a surrogate [40], [41]. Therefore, our model can be rewritten as

$$\begin{aligned} &\arg\min_{\mathbf{X},\mathbf{B},\mathbf{S}} \operatorname{Tr}\left(\mathbf{X}\mathbf{L}\mathbf{X}^T\right) + \lambda\|\mathbf{B}\|_1 + \beta\sum_{p=1}^{P}\|\mathbf{S}_p\|_* \\ &\text{s.t.} \quad \|\mathbf{Y}-\mathbf{X}-\mathbf{B}-\mathbf{S}\|_F^2 \le \xi, \|\mathbf{Y}_p-\mathbf{X}_p-\mathbf{B}_p-\mathbf{S}_p\|_F^2 \le \xi \\ &\qquad \operatorname{rank}(\mathbf{S}_p) \le r. \end{aligned} \tag{8}$$

D. Low Rankness Described by GLR

Generally, the low-rank property of HSI is modeled by the following equation [20], [28]:

$$\begin{aligned} &\min_{\mathbf{X}} \ \operatorname{rank}(\mathbf{X}) \\ &\text{s.t.} \quad \|\mathbf{Y}-\mathbf{X}\|_F^2 \le \xi, \operatorname{rank}(\mathbf{X}) \le r_1. \end{aligned} \tag{9}$$

As mentioned above, the low-rank constraint rank$(\mathbf{X})$ is non convex, so the nuclear norm $\|\mathbf{X}\|_*$ is used as a surrogate in the optimization step, which is formulated as

$$\begin{aligned} &\min \|\mathbf{X}\|_* \\ &\text{s.t.} \quad \operatorname{rank}(\mathbf{X}) \le r_1 \end{aligned} \tag{10}$$

where $\mathbf{X}$ can be recovered by the SVD, i.e., $\mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^T$. Meanwhile, as shown in (3), the GLR can be reformulated as follows:

$$\begin{aligned} \operatorname{Tr}\left(\mathbf{X}\mathbf{L}\mathbf{X}^T\right) &= \frac{1}{2}\sum_{i,j=1}^{K} W_{i,j}\left\|\mathbf{X}_i-\mathbf{X}_j\right\|_2^2 \\ &= \frac{1}{2}\sum_{i,j=1}^{K} W_{i,j}\left\|\left(\mathbf{U}_{\cdot i}-\mathbf{U}_{\cdot j}\right)\Sigma\mathbf{V}^T\right\|_2^2 \end{aligned} \tag{11}$$

where $\mathbf{U}_{\cdot i}$ is the ith row of matrix $\mathbf{U}$. Since Σ has no negative item, $\operatorname{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ tends to 0 when $\|\mathbf{X}\|_*$ is approaching 0,

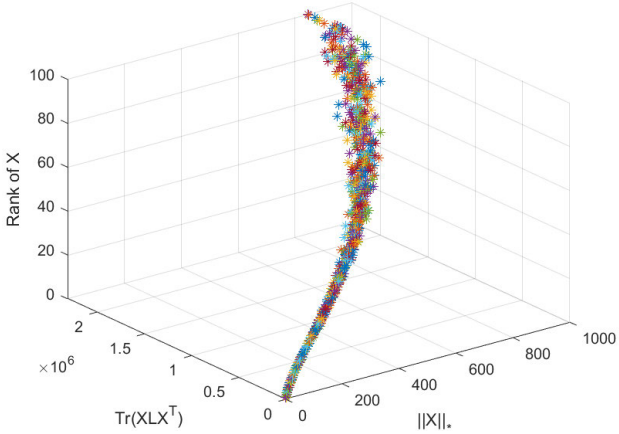


Fig. 3. Equivalence relation between $\operatorname{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ and $\|\mathbf{X}\|_*$, characterized by their monotonicity with rank$(\mathbf{X})$.

which means the minimization of $\operatorname{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ is similar to the minimization of $\|\mathbf{X}\|_*$.

To validate the above equivalence relation, we compute $\operatorname{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ and $\|\mathbf{X}\|_*$ of a random matrix $\mathbf{X}$ with $\operatorname{rank}(\mathbf{X}) \in [1, 100]$. As shown in Fig. 3, the value of $\operatorname{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ is monotonically decreasing when rank$(\mathbf{X})$ is approaching 1, so is the value of $\|\mathbf{X}\|_*$. In other words, our optimization model that minimizes the $\operatorname{Tr}(\mathbf{X}\mathbf{L}\mathbf{X}^T)$ is similar to the traditional model used in [20] and [28], but with more effective performance (more details in Section IV).

E. Optimization Procedure

The proposed FGLR model (8) can be solved by the augmented Lagrange multiplier (ALM) optimization procedure, which minimizes the following augmented Lagrangian function:

$$\begin{aligned} \{\mathbf{X},\mathbf{B},\mathbf{S}\} = \arg\min_{\mathbf{X},\mathbf{B},\mathbf{S}} &\operatorname{Tr}\left(\mathbf{X}\mathbf{L}\mathbf{X}^T\right) + \lambda\|\mathbf{B}\|_1 \\ &+ \mathcal{I}^X(\mathbf{Y}-\mathbf{X}-\mathbf{B}-\mathbf{S}) \\ &+ \frac{\rho}{2}\|\mathbf{Y}-\mathbf{X}-\mathbf{B}-\mathbf{S}\|_F^2 + \beta\|\mathbf{S}_p\|_* \\ &+ \mathcal{I}^S\left(\mathbf{Y}_p-\mathbf{X}_p-\mathbf{B}_p-\mathbf{S}_p\right) \\ &+ \frac{\rho}{2}\|\mathbf{Y}_p-\mathbf{X}_p-\mathbf{B}_p-\mathbf{S}_p\|_F^2 \end{aligned} \tag{12}$$

where ρ is the penalty coefficient, and $(\mathcal{I}^X, \mathcal{I}^S) \in \mathcal{I}$ is the Lagrange multiplier. Solving the above problems often involves optimizing a single variable while keeping other

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variables constant. Therefore, the items in (8) can be updated in an alternate iterative manner as follows:

$$\begin{aligned} X^{k+1} &= \arg\min \mathcal{L}\left(X, B^k, S_p^k, \mathcal{I}^k\right) \\ B^{k+1} &= \arg\min \mathcal{L}\left(X^{k+1}, B, S_p^k, \mathcal{I}^k\right) \\ S_p^{k+1} &= \arg\min \mathcal{L}\left(X_p^{k+1}, B_p^{k+1}, S_p, \mathcal{I}^k\right) \\ \mathcal{I}^{k+1} &= \mathcal{I}^k + \rho \cdot \left(Y - X^{k+1} - B^{k+1} - S^{k+1}\right). \end{aligned} \tag{13}$$

1) Optimization Regarding $\mathbf{X}$*:* For the optimization of the GLR, we refer to the solution in [33] and give the following explicit solution:

$$\begin{aligned} \mathbf{X} &= \arg\min_{\mathbf{X}} \operatorname{Tr}\left(\mathbf{X}\mathbf{L}\mathbf{X}^T\right) + \mathcal{I}^X(\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S}) \\ &\quad + \frac{\rho}{2}\|\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S}\|_F^2 \\ &= \arg\min_{\mathbf{X}} \operatorname{Tr}\left(\mathbf{X}\mathbf{L}\mathbf{X}^T\right) + \frac{\rho}{2}\left\|\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S} + \frac{\mathcal{I}^X}{\rho}\right\|_F^2 \\ &= \left(2\mathbf{L}^T + \rho\right)^{-1} \cdot \left(\rho(\mathbf{Y} - \mathbf{B} - \mathbf{S}) + \mathcal{I}^X\right). \end{aligned} \tag{14}$$

2) Optimization Regarding $\mathbf{B}$*:* Upon obtaining the updated component $\mathbf{X}$, the sparse noise $\mathbf{B}$ can be solved through a soft threshold operation

$$\begin{aligned} \mathbf{B} &= \arg\min_{\mathbf{B}} \|\mathbf{B}\|_1 + \mathcal{I}^X \cdot (\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S}) \\ &\quad + \frac{\rho}{2}\|\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S}\|_F^2 \\ &= \arg\min_{\mathbf{B}} \|\mathbf{B}\|_1 + \frac{\rho}{2}\left\|\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S} + \frac{\mathcal{I}^X}{\rho}\right\|_F^2 \\ &= \text{shrink_}\ell_1\left(\mathbf{Y} - \mathbf{X} - \mathbf{S} + \frac{\mathcal{I}^X}{\rho}, \frac{\lambda}{\rho}\right) \end{aligned} \tag{15}$$

where shrink_$\ell_1(\cdot)$ is a soft threshold operation [42], defined as follows:

$$\text{shrink_}\ell_1(x, \gamma) = \begin{cases} x - \gamma, & \text{if } x > \gamma \\ x + \gamma, & \text{if } x < \gamma \\ 0, & \text{else.} \end{cases} \tag{16}$$

3) Optimization Regarding $\mathbf{S}_p$*:* As mentioned above, the low-rank property of stripe noise is essentially a problem of low-rank matrix approximation (LRMA). Its augmented Lagrangian function is expressed as follows:

$$\begin{aligned} &\mathbf{S}_p \\ &= \arg\min_{\mathbf{S}_p} \sum_{b=1}^{P} \left(\beta\|\mathbf{S}_p\|_* + \mathcal{I}^S \cdot \left(\mathbf{Y}_p - \mathbf{X}_p - \mathbf{B}_p - \mathbf{S}_p\right)\right. \\ &\qquad \left. + \frac{\rho}{2}\|\mathbf{Y}_p - \mathbf{X}_p - \mathbf{B}_p - \mathbf{S}_p\|_F^2\right) \\ &= \arg\min_{\mathbf{S}_p} \sum_{b=1}^{P} \left(\beta\|\mathbf{S}_p\|_* + \frac{\rho}{2}\left\|\mathbf{Y}_p - \mathbf{X}_p - \mathbf{B}_p - \mathbf{S}_p + \frac{\mathcal{I}^S}{\rho}\right\|_F^2\right). \end{aligned} \tag{17}$$

For the LRMA problem, it can be solved by the SVD [42]. Therefore, the solution of stripe noise can be expressed as follows:

$$\begin{cases} \mathbf{S}_p = U\left(\text{shrink_}L_*\left(\Sigma_r, \frac{\beta}{\rho}\right)\right)V^* \\ \text{SVD}\left(\mathbf{Y}_p - \mathbf{X}_p - \mathbf{B}_p + \frac{\mathcal{I}^S}{\rho}\right) = U\Sigma V^* \\ \text{shrink_}L_*\left(\Sigma_r, \frac{\beta}{\rho}\right) = \text{diag}\left\{\max\left(\Sigma_{\text{rii}} - \frac{\beta}{\rho}, 0\right)\right\} \\ \Sigma_r = \text{diag}(\sigma_i(1 < i < r)). \end{cases} \tag{18}$$

After solving the three variables $\mathbf{X}$, $\mathbf{B}$ and $\mathbf{S}$, the Lagrange multiplier $\mathcal{I}^X$ and $\mathcal{I}^S$ can be updated in parallel as follows:

$$\begin{cases} \mathcal{I}^X = \mathcal{I}^X + \rho \cdot (\mathbf{Y} - \mathbf{X} - \mathbf{B} - \mathbf{S}) \\ \mathcal{I}^S = \mathcal{I}^S + \rho \cdot \left(\mathbf{Y}_p - \mathbf{X}_p - \mathbf{B}_p - \mathbf{S}_p\right). \end{cases} \tag{19}$$

The whole procedure is summarized in Algorithm 1.

Algorithm 1 FGLR

1: **Input**: observed HSI $\mathbf{Y}$,
stopping criterion $\zeta = 10^{-6}$, parameters λ, β, ε, r
2: **Initialize**;
$\mathbf{Y} = \mathbf{X}$; $\mathbf{S} = \mathbf{B} = 0$;
3: **for** iter=1:IterMax **do**
Update $\mathbf{X}$ via (14);
Solve (15) for $\mathbf{B}$;
Compute $\mathbf{S}_p$ by solving (18);
Update the Lagrangian multipliers in (19).
Check the convergence conditions.
$\|\mathbf{Y} - \mathbf{X} - \mathbf{S} - \mathbf{B}\|_F^2 \le \zeta$,
$\|\mathbf{Y_p} - \mathbf{X_p} - \mathbf{S_p} - \mathbf{B_p}\|_F^2 \le \zeta$,
end for
4: **Output**: Clean Image $\mathbf{X}$.

IV. Experimentation

A. Experiment Setup

1) Datasets: We conduct our experiments on simulated and real noisy HSI images, respectively. Two noise-free HSI datasets and two real noisy datasets are used. Their corresponding descriptions are as follows.

1) *ROSIS Pavia City Center Dataset:* This dataset is obtained by the reflection optical system imaging spectrometer (ROSIS-03) sensor. It contains 1096 × 1096 pixels and 102 bands, but some bands contain no information and should be discarded. A sub-image of 200 × 200 × 80 in [27] is selected.
2) *AVIRIS Cuprite:* This dataset is collected by the airborne visible/infrared imaging spectrometer sensor (AVIRIS) and the image contains 512 × 614 pixels and 224 spectral bands. Usually, 188 spectral bands with a high signal-to-noise ratio (SNR) are reserved as noiseless reference, and a sub-image of 200 × 200 × 188 is used in our experiments [43].
3) *Gaofen 5 (GF-5) Shanghai and Baoqing:* These two datasets are hyperspectral images taken by the GF-5 satellite in Shanghai and Baoqing, China. Both of them are corrupted by dense deadlines, Gaussian noise, and stripes. The image size is 300 × 300 × 155 [31].

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2) Comparison Methods: To validate the superiority of the proposed method, we compare the experimental results of FGLR with the 11 classical and state-of-the-art denoising methods, where seven of them are low-rank-based: LRMR [20], LLRSSTV [23], LRTDGS [24], RhyDe [26], L1HyMixDe [25], SSLR-SSTV [27], Double low-rank recovery (DLR) [28]. In addition, three graph-based methods and one deep learning-based method are considered, including the LRAWGTV [36], GSSTV [37], SRGLR [38]), and HSI-SDeCNN [30]. All of the methods are running on MATLAB R2018b (Intel[1] Core[2] i9-10900 CPU at 2.80 GHz, 128 GB RAM).

3) Evaluation Metrics: To quantitatively evaluate the denoising effect of different methods, we choose three common evaluation indicators, i.e., the peak SNR (PSNR) [44], [45], the structural similarity (SSIM) [46], and the spectral angle mapper (SAM) [47]. PSNR is the ratio between the power of the maximum intensity of an image and the noise power which is reflected by the mean square error. SSIM is an indicator that measures the SSIM between the restored image and the original image. SAM computes the spectral similarity by the angular difference between the spectrum vectors of the filtered and the noise-free HSI. We use the mean value of these indicators along the spectral dimension. The three evaluation indicators are defined as follows:

$$\begin{aligned}
\text{MPSNR} &= \frac{1}{b}\sum_{b=1}^{P} 10\log_{10}\frac{{I_{\max,b}}^2 \times MN}{\|\hat{u}_b - u_b\|^2} \\
\text{MSSIM} &= \frac{1}{b}\sum_{b=1}^{P} \frac{\left(2\mu_{u_b}\mu_{\hat{u}_b} + C_1\right)\left(2\sigma_{u_b\hat{u}_b} + C_2\right)}{\left(\mu_{u_b}^2 + \mu_{\hat{u}_b}^2 + C_1\right)\left(\sigma_{u_b}^2 + \sigma_{\hat{u}_b}^2 + C_2\right)} \\
\text{MSAM} &= \frac{1}{P}\sum_{b=1}^{P} \arccos\left[\frac{\sum_i^{MN} u_b^i \hat{u}_b^i}{\sqrt{\sum_{i=1}^{MN}\left(u_b^i\right)^2}\sqrt{\sum_{i=1}^{MN}\left(\hat{u}_b^i\right)^2}}\right]
\end{aligned} \tag{20}$$

where $I_{\max,b}$ refers to the maximum value of the image on the bth band. u_b and $\hat{u}_b$ represent clean HSI and denoising result on bth band, respectively. μ_{u_b} and $\mu_{\hat{u}_b}$ are average values of clean HSI u_b and denoised HSI $\hat{u}_b$, and σ_{u_b}, $\sigma_{\hat{u}_b}$ denote the the variances, while $\sigma_{u_b\hat{u}_b}$ represents the covariance between two HSI data.

4) Simulation Configurations: Both Pavia City Center dataset and Cuprite dataset are noise-free HSIs, and the pixel value has been normalized to [0, 1]. We simulate the noisy HSIs by adding a mixture of Gaussian noise, sparse noise, and stripe noise. Gaussian and sparse noise are added to each band, stripe noise is added to 30% randomly selected bands. The noise settings are described as follows.

Case 1: Every band is contaminated by i.i.d. Gaussian noise with standard variance $\sigma = 0.05$, and an impulse noise with the percentage of $o = 0.01$. In addition, for each band with stripe noise, $r = 30\%$ of the columns are corrupted by stripes, with an intensity $v = 0.075$.

[1]Registered trademark.
[2]Trademarked.

Case 2: The Gaussian noise and impulse noise are as same as in Case 1. Besides, the percentage and intensity of stripe noise are $r = 50\%$ and $v = 0.075$, respectively.

Case 3: The Gaussian noise and impulse noise are as same as in Case 1. The percentage and intensity of stripe noise are $r = 70\%$ and $v = 0.075$, respectively.

Case 4: The Gaussian noise and impulse noise are as same as in Case 1. Moreover, the percentage and intensity of stripe noise are $r = 30\%$ and $v = 0.05$, respectively.

Case 5: The Gaussian noise and impulse noise are as same as in Case 1. The percentage and intensity of stripe noise are $r = 30\%$ and $v = 0.1$, respectively.

Case 6: The random Gaussian noise with standard variance $\sigma = 0$–0.1 and random impulse noise with the percentage of $o = 0$–0.02 are added to each band. Furthermore, the percentage and intensity of stripe noise are $r = 50\%$ and $v = 0.1$, respectively.

Case 7: The Gaussian noise and impulse noise are simulated in the same way as in Case 6. The percentage and intensity of stripe noise are $r = 70\%$ and $v = 0.1$, respectively.

Case 8: The Gaussian noise and impulse noise are the same as in Case 1. Besides, wide vertical stripes are added, of which the width is 30% of M, i.e., 60 pixels. Each wide stripe starts at the 100 column with the intensity $v = 0.075$.

5) Parameters Setting: In the simulation experiment and the real experiment, we set up, respectively, $\lambda = 0.21$, $\beta = 3.7$, which are analyzed in Section IV-D2.

B. Simulated Datasets Experiments

1) Visual Effect: Figs. 4 and 5 show the denoising results by different methods on the 19th and 50th band of Pavia City Center and Cuprite data under Case 3, respectively. From the results, we can see that all of the methods can remove the noises more or less. Apparently, HSI-SDeCNN, SSLRSSTV, DLR, SRGLR, and FGLR have better visual performance. This is because HSI-SDeCNN uses the mixture noise under different cases to train the model, and the other four methods consider the stripe noise as an independent component and constrain it by using a low-rank term. Furthermore, it can be seen from the enlarged image of Fig. 5 that SSLRSSTV, DLR, and the proposed method can preserve more details and are more approximate to the original image.

In addition, Fig. 6 shows the results of different methods in removing the wide stripe noise in Case 8. Obviously, SSLRSSTV and FGLR have better visual effects when removing wide stripe noise. In terms of the recovered details, FGLR has a better performance.

2) Quantitative Results: Tables I and II present the quantitative results including the MPSNR, MSSIM, MSAM, and time cost of different methods under different noisy cases. The optimal values are in bold, and the second best are underlined. The MPSNR of the proposed method surpasses the other methods in most of the cases, and the MSSIM and MSAM rank first or second in most of the cases.

3) Mean Vertical Profiles: Figs. 7 and 8 show the mean vertical profiles of the 19th band of the Pavia City Center data and the 50th band of the Cuprite data, respectively. In these two figures, the horizontal axis represents column index and the

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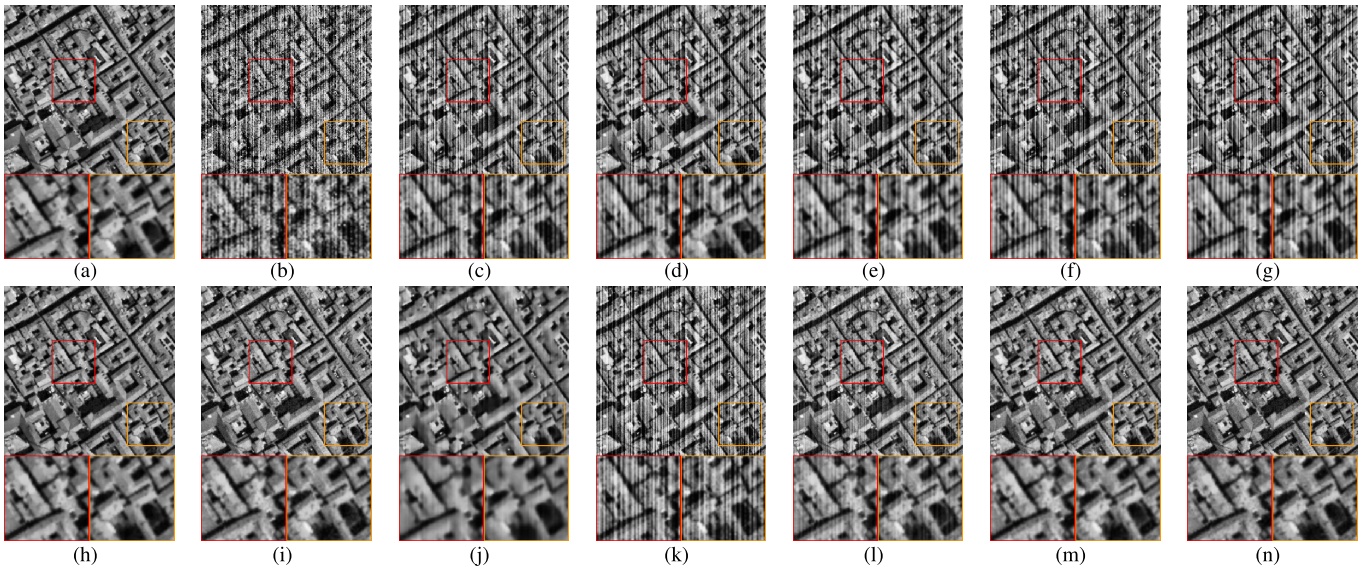


Fig. 4. Restoration results for Pavia image (Case 3) in bands 19 via the different methods. (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

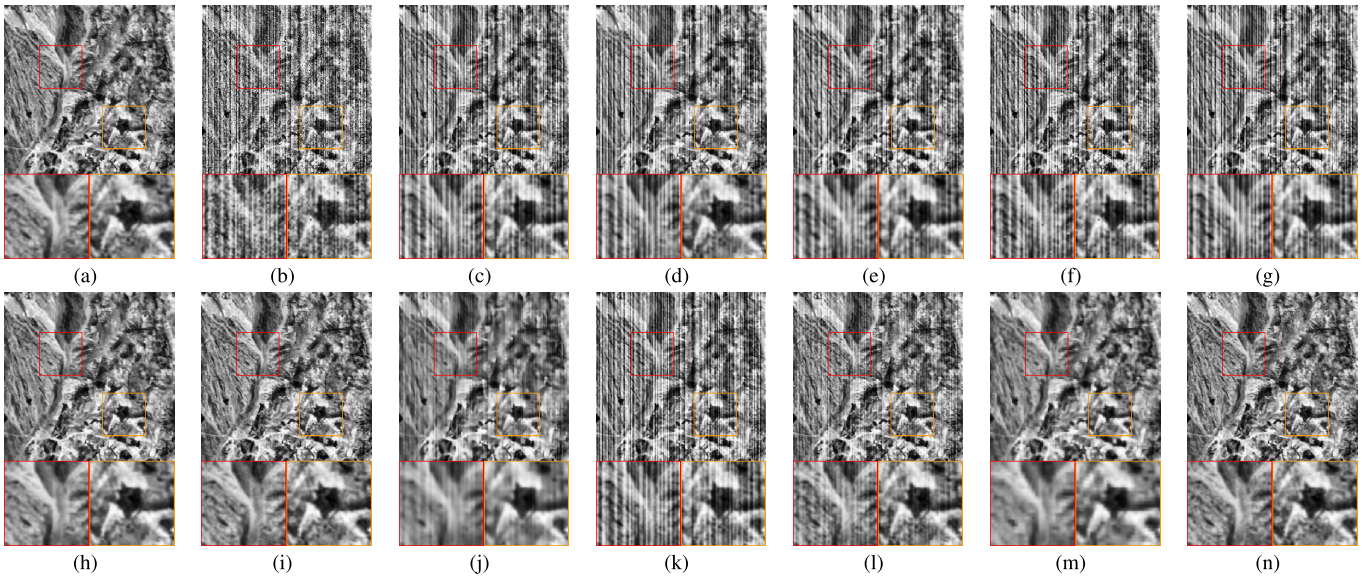


Fig. 5. Restoration results for Cuprite image (Case 3) in bands 50 via the different methods. (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

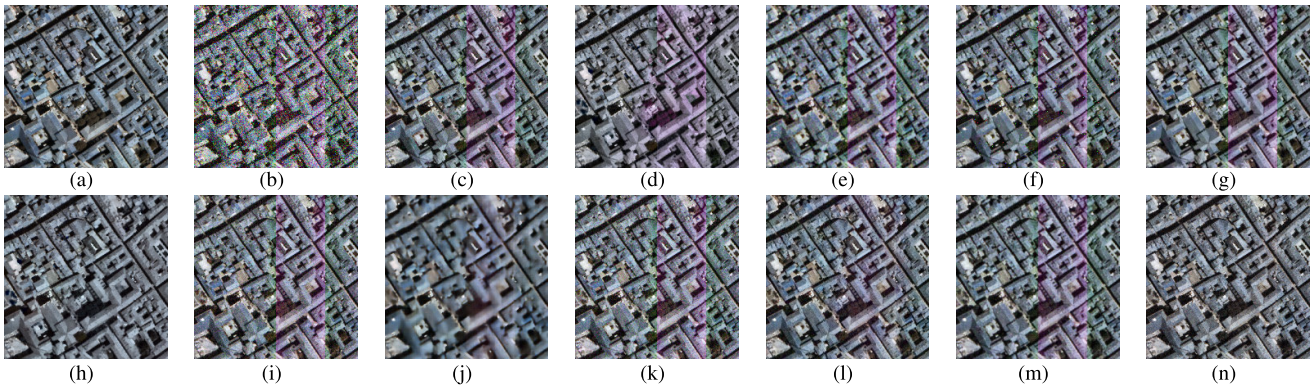


Fig. 6. Restoration results for Pavia image (Case 8) in bands (7, 14, 20) via the different methods. (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

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TABLE I

IN THE DIFFERENT CASES OF SIMULATED MIXED NOISE (CASES 1–8), THE QUANTITATIVE PICTURE QUALITY INDICES OF DENOISING THE PAVIA CITY CENTER DATA 200 × 200 × 80 WITH DIFFERENT METHODS

Case	Index	Noisy	LRMR	LLRSSTV	LRTDGS	RhyDe	L1HyMixDe	SSLRSSTV	DLR	HSI-SDeCNN	LRAWGTV	GSSTV	SRGLR	Proposed
Case 1 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.075$	MPSNR	21.9965	34.2711	34.2711	33.4876	31.4928	35.3876	35.9404	36.4492	32.4234	33.3452	35.6262	36.0600	**36.9344**
	MSSIM	0.6171	0.9231	0.9231	0.9190	0.8941	0.9240	0.9684	0.9599	0.9099	0.9022	0.9575	0.9649	**0.9690**
	MSAM	0.4065	0.1347	0.1347	0.1404	0.1513	0.1360	0.0835	0.1083	0.0837	0.1678	0.0779	**0.0725**	0.0833
	Time(S)	—	18.7	68.9	9.5	47.8	132.4	91.6	22.0	1.4	667.0	815.2	199.6	2.2
Case 2 $\sigma = 0.05$, $o = 0.01$, $r = 50\%$, $v = 0.075$	MPSNR	21.8069	33.5066	31.1852	32.9356	31.0069	34.9842	35.7635	33.9079	32.2080	33.6946	34.3402	36.0034	**36.9523**
	MSSIM	0.6035	0.8956	0.9006	0.8990	0.8710	0.9017	0.9675	0.9296	0.9602	0.9025	0.9450	0.9640	**0.9708**
	MSAM	0.4162	0.1702	0.1601	0.1667	0.1808	0.1713	0.0820	0.1477	0.0824	0.1802	0.0783	**0.0737**	0.0805
	Time(S)	—	18.7	68.5	9.4	48.9	130.8	95.2	21.9	1.4	678.8	884.6	213.1	2.0
Case 3 $\sigma = 0.05$, $o = 0.01$, $r = 70\%$, $v = 0.075$	MPSNR	21.6737	33.0097	29.9663	32.4143	30.5087	34.3487	35.8176	**37.5898**	32.0031	32.9063	33.2685	36.0308	37.2341
	MSSIM	0.5920	0.8688	0.8600	0.8751	0.8447	0.8756	0.9674	0.9672	0.9014	0.8809	0.9272	0.9637	**0.9708**
	MSAM	0.4277	0.2095	0.2066	0.1988	0.2188	0.2169	0.0814	0.0963	0.0837	0.2032	0.0849	0.0746	**0.0719**
	Time(S)	—	18.6	68.4	9.6	47.6	131.7	100.5	22.0	1.4	680.8	945.2	198.9	2.0
Case 4 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.05$	MPSNR	22.1778	35.3324	35.3313	34.1498	32.2773	36.4527	36.0780	**36.8964**	32.8417	35.1587	36.8933	35.9326	36.6573
	MSSIM	0.6282	0.9509	0.9619	0.9385	0.9190	0.9521	0.9688	0.9631	0.9149	0.9412	0.9683	0.9647	**0.9699**
	MSAM	0.3978	0.1058	0.0879	0.1215	0.1301	0.1095	0.0845	0.1000	0.0795	0.1311	**0.0715**	0.0719	0.0798
	Time(S)	—	18.7	68.6	9.6	48.6	132.5	95.4	22.0	1.4	669.6	771.0	197.3	1.9
Case 5 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.1$	MPSNR	21.8084	33.3412	33.1497	32.7234	30.8251	34.6309	35.6895	**37.4931**	32.0973	33.7168	34.1461	35.9098	37.1698
	MSSIM	0.6042	0.8959	0.8969	0.8979	0.8704	0.9012	0.9675	0.9644	0.9056	0.9102	0.9447	0.9636	**0.9709**
	MSAM	0.4158	0.1590	0.1564	0.1580	0.1709	0.1534	0.0821	0.0982	0.0837	0.1653	0.0831	**0.0736**	0.0795
	Time(S)	—	18.7	68.5	9.4	47.9	132.9	95.3	22.0	1.4	678.8	886.5	197.3	2.1
Case 6 $\sigma = 0 \sim 0.1$, $o = 0 \sim 0.02$, $r = 50\%$, $v = 0.1$	MPSNR	22.0504	31.6529	29.2427	31.8767	30.8410	36.3710	36.2651	35.2258	31.6871	34.8828	32.1820	34.9981	**36.6189**
	MSSIM	0.5908	0.8440	0.8443	0.8569	0.8418	0.8727	**0.9715**	0.8974	0.8974	0.9045	0.9110	0.9584	0.9655
	MSAM	0.4403	0.2173	0.2189	0.2133	0.2129	0.1972	**0.0784**	0.1844	0.0816	0.1862	0.0873	0.0799	0.0836
	Time(S)	—	17.1	66.1	9.6	47.5	129.6	90.8	22.1	1.4	673.7	898.4	198.7	2.0
Case 7 $\sigma = 0 \sim 0.1$, $o = 0 \sim 0.02$, $r = 70\%$, $v = 0.1$	MPSNR	21.6457	30.9703	28.0123	31.4302	30.3095	36.1353	36.0715	35.9125	30.8157	35.9597	31.0055	34.6611	**36.1567**
	MSSIM	0.4707	0.8190	0.8072	0.8362	0.8221	0.8555	**0.9700**	0.9157	0.8782	0.9234	0.8898	0.9550	0.9606
	MSAM	0.5633	0.2631	0.2769	0.2462	0.2573	0.2524	**0.0806**	0.1922	0.0917	0.1818	0.0986	0.0815	0.0875
	Time(S)	—	19.6	65.0	10.2	48.6	126.5	87.1	22.1	1.4	664.4	1213.8	215.0	2.1
Case 8 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.075$	MPSNR	21.9814	34.0935	32.9269	33.0917	31.4034	35.4414	34.2579	34.4311	31.4509	34.3031	34.9235	32.7651	**35.5445**
	MSSIM	0.6325	0.9593	0.9571	0.9499	0.9338	**0.9689**	0.9653	0.9540	0.9079	0.9498	0.9682	0.9520	0.9667
	MSAM	0.3880	0.1191	0.1139	0.1318	0.1365	0.1154	0.1200	0.1319	**0.0945**	0.1631	0.1090	0.1112	0.1086
	Time(S)	—	30.9	78.0	9.8	49.6	137.5	104.8	22.0	2.8	680.4	724.2	197.5	2.0

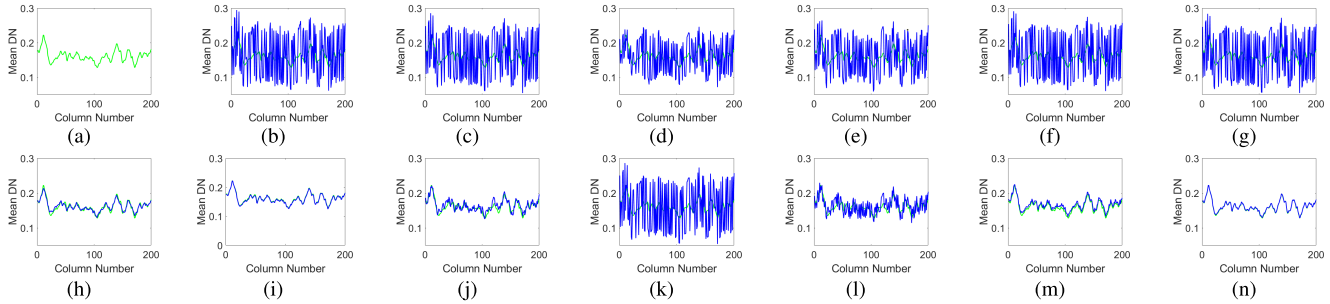


Fig. 7. Mean vertical profiles of band 19 in the Pavia city center image (Case 3). (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

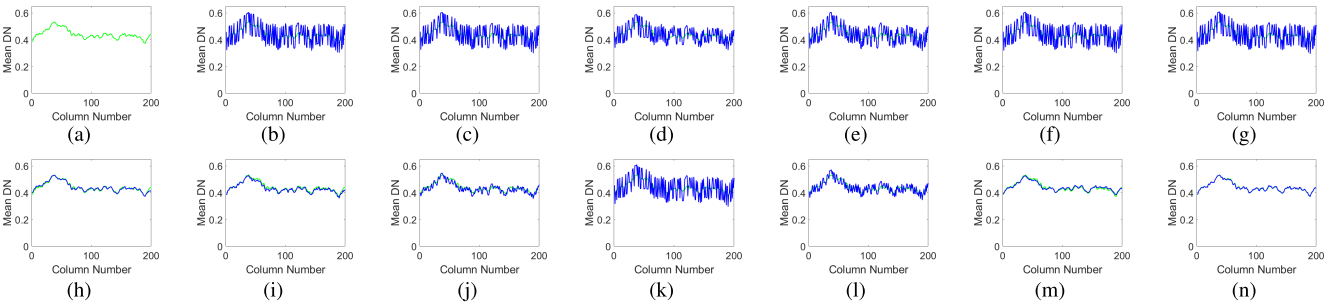


Fig. 8. Mean vertical profiles of band 50 in the Cuprite image (Case 3). (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

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TABLE II

IN THE DIFFERENT CASES OF SIMULATED MIXED NOISE (CASES 1–8), THE QUANTITATIVE PICTURE QUALITY INDICES OF DENOISING THE CUPRITE DATA 200 × 200 × 188 WITH DIFFERENT METHODS

Case	Index	Noisy	LRMR	LLRSSTV	LRTDGS	RhyDe	L1HyMixDe	SSLRSSTV	DLR	HSI-SDeCNN	LRAWGTV	GSSTV	SRGLR	Proposed
Case 1 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.075$	MPSNR	22.5151	34.5463	32.3528	34.5343	30.4660	35.5361	34.2927	33.0609	31.4418	35.3693	34.6338	30.2502	**36.4655**
	MSSIM	0.5078	0.8882	0.9018	0.8881	0.8152	0.8859	0.9426	0.9416	0.8413	0.9036	0.9146	0.8414	**0.9485**
	MSAM	0.1656	0.0457	0.0488	0.0421	0.0639	0.0455	0.0437	0.0543	0.0364	0.0543	**0.0345**	0.0530	0.0369
	Time(S)	—	31.0	131.1	16.9	46.6	115.8	170.9	69.0	3.4	1648.9	2668.3	361.7	4.4
Case 2 $\sigma = 0.05$, $o = 0.01$, $r = 50\%$, $v = 0.075$	MPSNR	22.3306	34.0142	30.5854	34.1388	30.0053	35.1019	34.2685	33.0905	31.4022	34.3778	33.3327	30.2846	**36.5943**
	MSSIM	0.4958	0.8523	0.8596	0.8618	0.7867	0.8543	0.9427	0.9408	0.8400	0.8758	0.8874	0.8468	**0.9465**
	MSAM	0.1695	0.0578	0.0607	0.0514	0.0736	0.0572	0.0438	0.0540	**0.0366**	0.0613	0.0377	0.0541	0.0373
	Time(S)	—	30.7	130.0	18.0	46.6	119.0	170.5	70.3	3.4	1659.2	2668.1	365.5	4.5
Case 3 $\sigma = 0.05$, $o = 0.01$, $r = 70\%$, $v = 0.075$	MPSNR	22.1512	33.4173	29.2822	33.6028	29.5805	34.5467	34.2701	33.0450	31.2210	34.4961	32.1230	30.2904	**36.5259**
	MSSIM	0.4846	0.8257	0.8173	0.8376	0.7647	0.8295	0.9426	0.9421	0.8335	0.8710	0.8625	0.8487	**0.9486**
	MSAM	0.1731	0.0685	0.0747	0.0592	0.0822	0.0677	0.0437	0.0545	**0.0366**	0.0657	0.0391	0.0546	0.0370
	Time(S)	—	31.1	129.6	17.9	46.6	119.9	171.1	69.9	3.4	1647.7	2668.5	369.3	4.3
Case 4 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.05$	MPSNR	22.7128	35.8942	34.0531	35.5465	31.1950	**36.7263**	34.3309	33.1435	31.6466	35.1139	36.4715	30.3199	36.6234
	MSSIM	0.5187	0.9262	0.9362	0.9160	0.8433	0.9202	0.9425	0.9423	0.8450	0.9044	0.9363	0.8429	**0.9472**
	MSAM	0.1626	0.0371	0.0424	0.0361	0.0581	0.0383	0.0436	0.0546	0.0362	0.0526	**0.0326**	0.0526	0.0371
	Time(S)	—	31.0	131.8	16.8	46.3	119.3	171.9	73.2	3.4	1679.8	2668.4	356.1	4.0
Case 5 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.1$	MPSNR	22.3035	33.9404	30.4973	34.0183	29.9422	35.0599	34.2659	33.0308	31.1847	34.3436	33.1483	30.1291	**36.5812**
	MSSIM	0.4963	0.8569	0.8605	0.8636	0.7904	0.8592	0.9426	0.9418	0.8364	0.8791	0.8890	0.8394	**0.9482**
	MSAM	0.1696	0.0549	0.0612	0.0492	0.0712	0.0534	0.0438	0.0552	**0.0372**	0.0602	0.0389	0.0543	0.0373
	Time(S)	—	30.9	130.1	17.7	46.8	120.0	171.2	63.0	3.4	1654.2	2668.5	363.4	4.2
Case 6 $\sigma = 0 \sim 0.1$, $o = 0 \sim 0.02$, $r = 50\%$, $v = 0.1$	MPSNR	22.2572	32.5071	28.4933	33.3780	29.3716	35.9543	34.4283	33.2579	30.9348	34.5612	31.4723	30.3397	**36.1208**
	MSSIM	0.4937	0.8063	0.7935	0.8264	0.7527	0.8388	0.9476	**0.9502**	0.8271	0.8570	0.8480	0.8466	0.9409
	MSAM	0.1859	0.0731	0.0869	0.0629	0.0852	0.0634	0.0438	0.0575	**0.0386**	0.0722	0.0422	0.0536	0.0391
	Time(S)	—	56.4	145.1	18.6	45.4	28.4	191.2	62.5	4.4	1709.4	2674.4	379.2	4.2
Case 7 $\sigma = 0 \sim 0.1$, $o = 0 \sim 0.02$, $r = 70\%$, $v = 0.1$	MPSNR	21.9291	32.1942	27.8749	33.1497	29.0753	**36.0014**	34.5512	32.2751	30.6369	35.4583	30.4460	30.0825	35.9736
	MSSIM	0.4737	0.7835	0.7627	0.8050	0.7306	0.8186	**0.9476**	0.9331	0.8120	0.8676	0.8115	0.8428	0.9376
	MSAM	0.1955	0.0894	0.1036	0.0758	0.1003	0.0784	0.0435	0.0617	0.0402	0.0757	0.0461	0.0560	**0.0399**
	Time(S)	—	57.6	145.9	18.6	46.3	126.7	191.4	61.2	3.3	1694.6	2668.9	384.0	4.1
Case 8 $\sigma = 0.05$, $o = 0.01$, $r = 30\%$, $v = 0.075$	MPSNR	22.5223	34.8164	32.3085	34.3320	30.4808	**35.8353**	32.9334	32.4792	31.0252	35.3386	34.8075	29.5527	35.0624
	MSSIM	0.5283	0.9502	0.9432	0.9408	0.8715	**0.9556**	0.9416	0.9439	0.8460	0.9363	0.9488	0.8399	0.9499
	MSAM	0.1627	0.0452	0.0519	0.0438	0.0627	0.0436	0.0504	0.0611	**0.0433**	0.0522	0.0446	0.0586	0.0439
	Time(S)	—	38.3	140.6	17.1	48.2	118.4	176.4	59.7	3.3	1638.3	2365.9	363.5	4.1

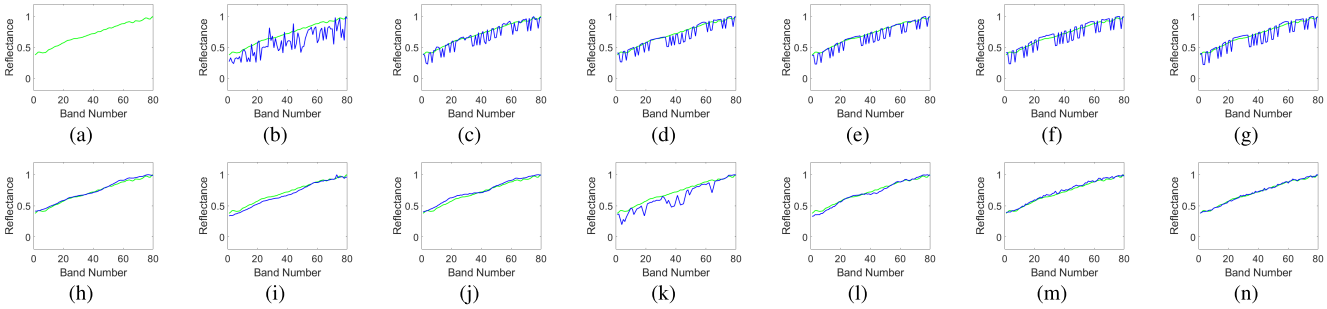


Fig. 9. Original spectrum (green curve) and the denoising results (blue curve) of pixel (116, 135) in simulated Case 3 of the Pavia city center image. (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

vertical axis represents the mean digital number value of each column. The green curve is the mean vertical profiles of the clean HSI, and the blue curve represents either the noisy HSI or the denoised images by different methods. It is clear that the curves obtained by LRMR, LLRSSTV, LRTDGS, RhyDe, L1HyMixDe, LRAWGTV, and GSSTV are very catastrophic due to the presence of stripe noise, while the curves of HSI-SDeCNN, SSLRSSTV, DLR, SRGLR, and FGLR are close to clean HSI. Moreover, by careful comparison, it can be found that the mean vertical profiles obtained by our FGLR on these two datasets are closer to the clean HSI than HSI-SDeCNN, SSLRSSTV, DLR, and SRGLR.

4) Spectrum Curves: To analyze the spectral recovery capabilities of each method, we compare the spectrum curves. Figs. 9 and 10 shows the original spectrum (green curve) and the denoised spectrums (blue curve) of the pixel (116, 135) of the Pavia City Center data and the pixel (183, 169) of the Cuprite data. As can be seen clearly, the spectrum curve recovered by FGLR is the most consistent with the original spectral curve, while the others are affected by the noise more or less. These curves further indicate the superiority and robustness of FGLR.

5) Overall PSNR and SSIM Curves of Each Band: Fig. 11 presents the PSNR and SSIM curves of Pavia City Center data

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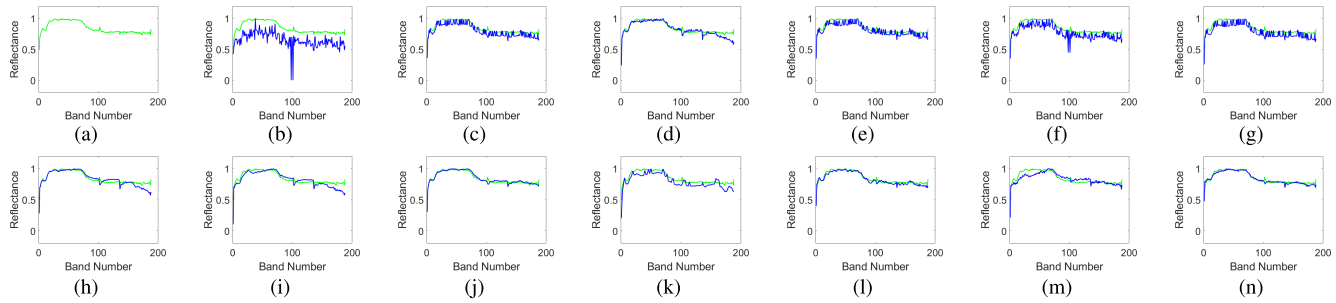


Fig. 10. Original spectrum (green curve) and the denoising results (blue curve) of pixel (183, 169) in simulated Case 3 of the Cuprite image. (a) Original image. (b) Noise image. (c) LRMR. (d) LLRSSTV. (e) LRTDGS. (f) RhyDe. (g) L1HyMixDe. (h) SSLRSSTV. (i) DLR. (j) HSI-SDeCNN. (k) LRAWGTV. (l) GSSTV. (m) SRGLR. (n) FGLR (proposed).

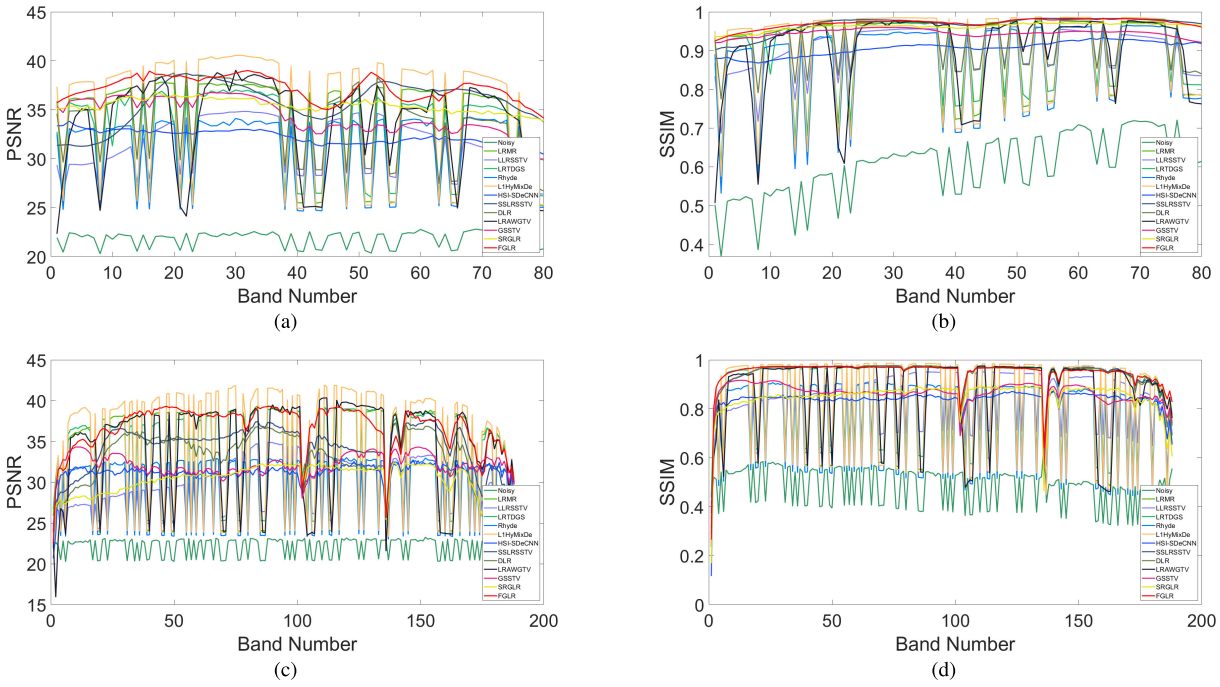


Fig. 11. Overall PSNR and SSIM values of the 12 competing methods. (a) and (b) Indexes of the Pavia city center dataset (Case 2). (c) and (d) Indexes of Cuprite dataset (Case 3).

and Cuprite data using different methods under Cases 2 and 3, respectively. As shown in Fig. 11, the PSNR and SSIM curves of most methods have obvious saw-tooth shapes on these bands, which is caused by the existence of stripe noise in several randomly selected bands. FGLR considers the stripe noise as an independent component and performs well with these stripe noise, so its curves are comparatively smoother. For both datasets, the PSNR and SSIM curves obtained by FGLR (red curve) are higher and more stable than the others in most of the bands. Despite the fact that the PSNR curve of L1HyMixDe is higher than FGLR in some bands, it is affected by the saw-tooth, so its mPSNR is lower than FGLR in most of the cases. In addition, although the PSNR and SSIM of the FGLR are not always the best among all the spectral bands, they can achieve comparable or better performance in the least time. These overall PSNR and SSIM curves further demonstrate the reliability and effectiveness of our algorithm.

6) Classification Results of Different Denoising Methods: To verify the impact of the denoising results of different methods on the subsequent applications, we select the results of Pavia City Center data under Case 3 for HSI classification. Two classification algorithms are used, the first one is K nearest neighbors (KNN) [48] and the second one is 3DCNN [49]. Two evaluation metrics are considered: the overall classification accuracy (OA) and the kappa coefficients. Higher OA and kappa indicate better classification performance. Table III displays the corresponding classification results. Obviously, whether using KNN or 3DCNN for classification, the denoised image obtained from FGLR achieves the highest OA and kappa values, which also reflects the better denoising effect of FGLR.

C. Real Noisy Dataset Experiments

Fig. 12 shows the pseudo-color image of Shanghai dataset using band (38, 152, 155). It can be found that the original image is affected by a small amount of different noises, especially the stripe noise. After denoising, the results of LRMR, LRTDGS, LRAWGTV, and GSSTV still have obvious stripe noise. The result of LLRSSTV retains stripe noise in

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TABLE III
CLASSIFICATION ACCURACIES WITH DIFFERENT DENOISING ALGORITHM ON CASE 3 OF THE PAVIA CITY CENTER IMAGE

Method	Index	LRMR	LLRSSTV	LRTDGS	RhyDe	L1HyMixDe	SSLRSSTV	DLR	HSI-SDeCNN	LRAWGTV	GSSTV	SRGLR	FGLR(Proposed)
KNN	Accuracy	0.9861	0.9866	0.9840	0.9852	0.9868	0.9891	0.9841	0.9847	0.9843	0.9879	0.9899	**0.9931**
	Kappa	0.885	0.892	0.867	0.879	0.892	0.909	0.868	0.873	0.870	0.900	0.919	**0.944**
3DCNN	Accuracy	0.9738	0.9700	0.9834	0.9772	0.9822	0.9762	0.9819	0.9715	0.9791	0.9870	0.9700	**0.9881**
	Kappa	0.791	0.761	0.868	0.809	0.858	0.809	0.856	0.771	0.832	0.895	0.759	**0.905**

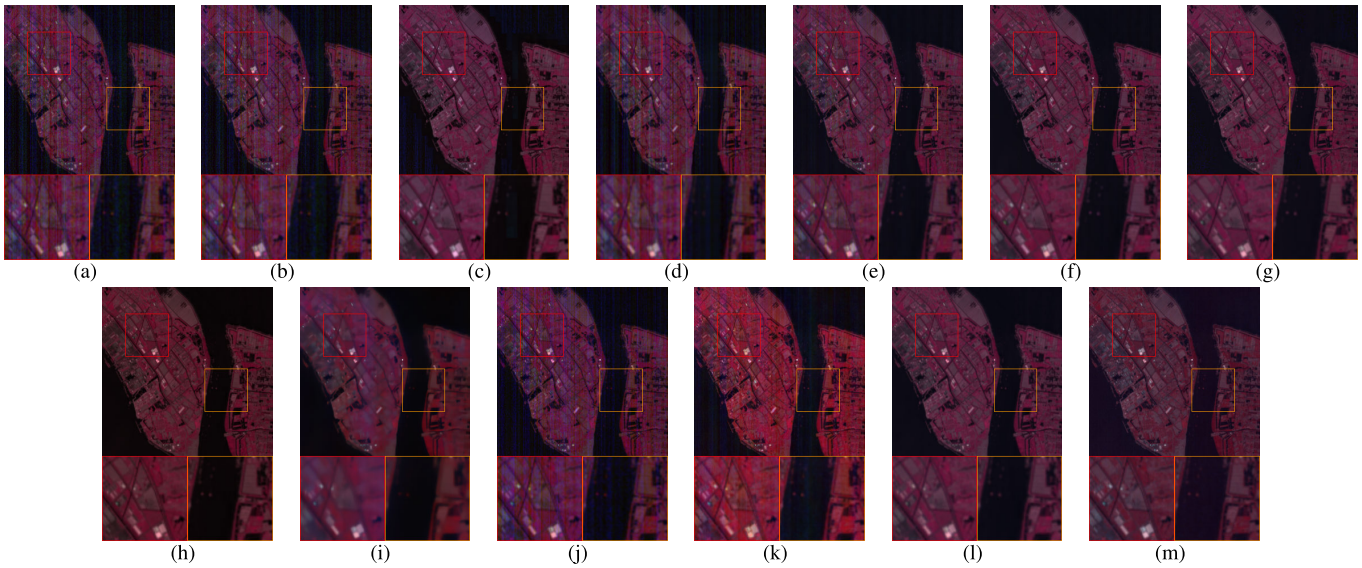


Fig. 12. Restoration results for GF-5 Shanghai image in bands (38, 152, 155) via the different methods. (a) Original image. (b) LRMR. (c) LLRSSTV. (d) LRTDGS. (e) RhyDe. (f) L1HyMixDe. (g) SSLRSSTV. (h) DLR. (i) HSI-SDeCNN. (j) LRAWGTV. (k) GSSTV. (l) SRGLR. (m) FGLR (proposed).

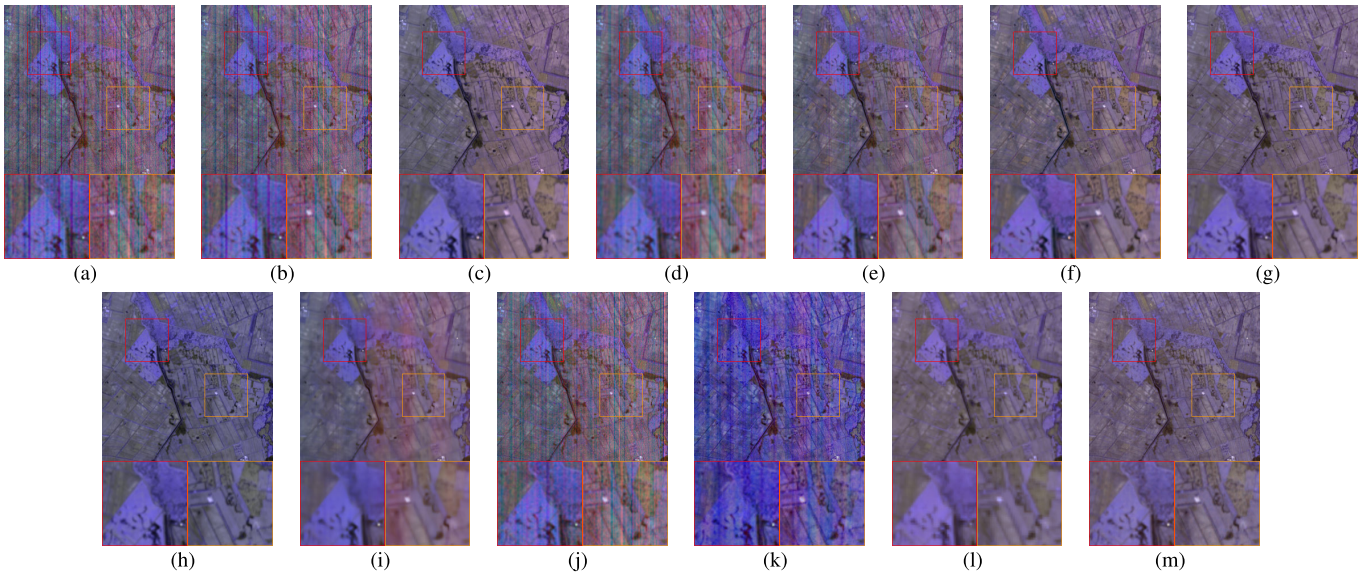


Fig. 13. Restoration results for GF-5 Baoqing image in bands (155, 152, 86) via the different methods. (a) Original image. (b) LRMR. (c) LLRSSTV. (d) LRTDGS. (e) RhyDe. (f) L1HyMixDe. (g) SSLRSSTV. (h) DLR. (i) HSI-SDeCNN. (j) LRAWGTV. (k) GSSTV. (l) SRGLR. (m) FGLR (proposed).

dark areas, and the result of HSI-SDeCNN is over smoothing. Due to that the noise level is not very high in this dataset, the other methods can yield good visual results, including RhyDe, L1HyMixDe, SSLRSSTV, DLR, SRGLR, and FGLR.

Fig. 13 depicts pseudo-color image of Baoqing dataset using (155, 152, 86). Compared with Fig. 12(a), Fig. 13(a) is more severely corrupted by stripes, deadlines, serious Gaussian noise, and impulse noise. The restored results of LRMR, LRTDGS, RhyDe, HSI-SDeCNN, LRAWGTV, and GSSTV still contain some Gaussian and stripe noise. SRGLR can remove noise well, but its result is oversmoothed. Among the denoising results, LLRSSTV, L1HyMixDe, SSLRSSTV, DLR, and FGLR have the best visual performance and preserve the global structure.

D. Discussion

1) Convergence of FGLR: Similar to [50], we use the value of $\|\mathbf{Y}-\mathbf{X}^{k+1}-\mathbf{B}^{k+1}-\mathbf{S}^{k+1}\|_{\infty}$ as a relative change to verify the convergence of our method. Fig. 14 shows the iteration

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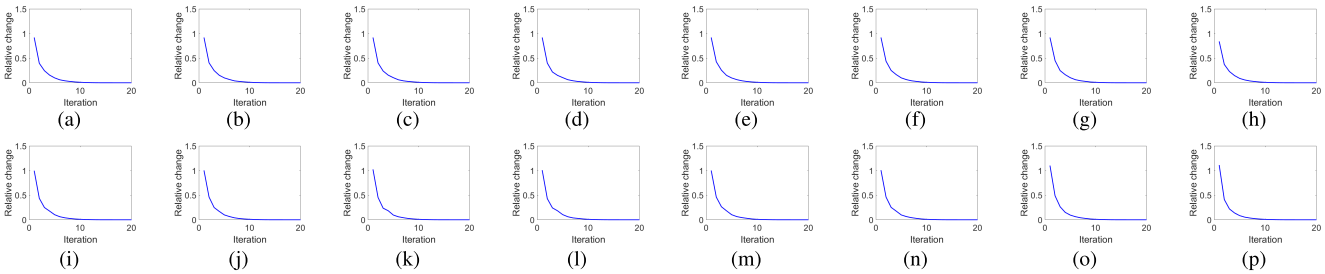


Fig. 14. (a)–(h) and (i)–(p) Relative change values of FGLR from Case 1–8 of the Pavia city center dataset and the Cuprite dataset, with respect to the iteration number.

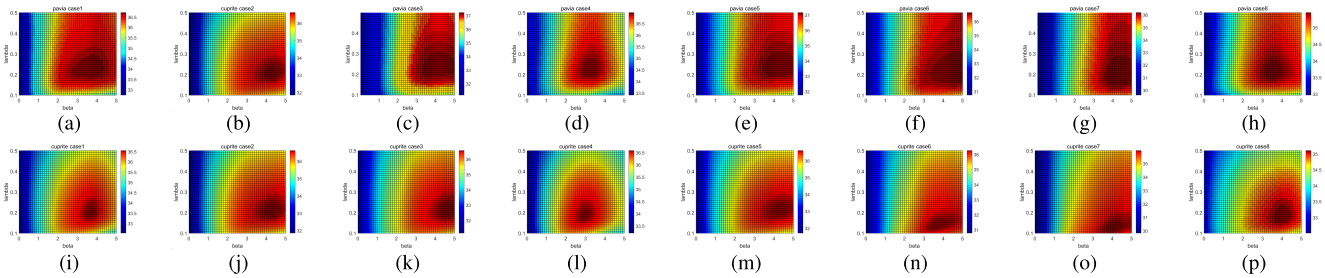


Fig. 15. (Left to right) MPSNR values of FGLR (proposed method) for (a)–(h) Pavia city center dataset and (i)–(p) Cuprite dataset by varying parameters λ and β. The data were corrupted by the noise simulated in Case 1–8.

TABLE IV
ABLATION STUDY OF DIFFERENT FGLR MODULES

modules	Index	Case 2	Case 3	Case 6	Case 8
$Tr(\mathbf{XLX}^T)$	MPSNR	32.1078	31.4753	24.0614	32.4864
	MSSIM	0.9119	0.8954	0.6714	0.9390
	MSAM	0.1107	0.1163	0.2310	0.1008
$Tr(\mathbf{XLX}^T)+\lambda\|B\|_1$	MPSNR	33.6070	32.7037	32.2567	34.6186
	MSSIM	0.9369	0.9194	0.9131	0.9663
	MSAM	0.0929	0.0997	0.1259	0.0822
FGLR with $\mathcal{I}^X = \mathcal{I}^Y = 0$	MPSNR	31.8487	31.5229	30.8481	32.2145
	MSSIM	0.9304	0.9224	0.9019	0.9567
	MSAM	0.1045	0.1053	0.1158	0.1043
FGLR	MPSNR	36.5943	37.2341	35.7520	35.5445
	MSSIM	0.9465	0.9708	0.9599	0.9667
	MSAM	0.0373	0.0805	0.0987	0.1086

curves of the Pavia City Center dataset and Cuprite dataset with the eight noise cases, respectively. It can be found that the relative change converges when the number of iterations exceeds 5.

2) Parameter Analysis: In the proposed FGLR optimization model (8), there are two parameters, namely λ and β. λ is used to control the sparse regularizer $\|\mathbf{B}\|_1$, and β is the coefficient for the low-rank term $\sum_{p=1}^{P}\|\mathbf{S}_p\|$. In Fig. 15, we show the MPSNR values of two simulated datasets under different cases, with different λ and β, where λ ranges from 0.1 to 0.5 and β ranges from 0 to 5. The first row of images represents the MPSNR values of Pavia City Center data, and the second row represents the MPSNR values of Cuprite data. As shown, most of the highest PSNRs are concentrated in a certain range. To make a compromise for all the cases, we set $\lambda = 0.21$ and $\beta = 3.7$ for both the simulated and real HSI data.

In addition, according to Section II-C, we set $r = 1$ for the upper bound rank of the stripe noise $\mathbf{S}_p$. For ε in (6), we use $\varepsilon = 50$ empirically.

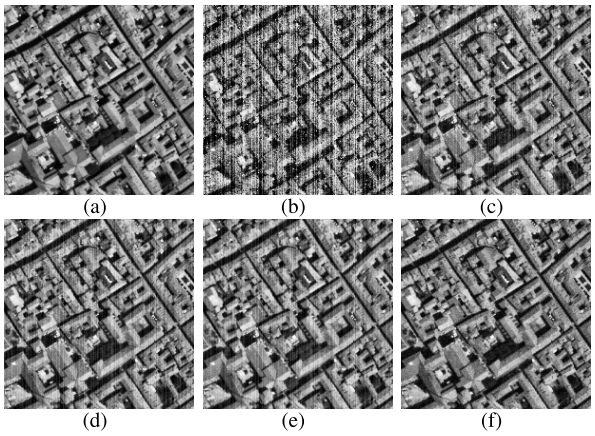


Fig. 16. (a) Original Pavia city center data (band 19). (b) Noise image under Case 2. (c)–(f) Denoising results obtained by different modules (c) Tr($\mathbf{XLX}^T$), (d) Tr($\mathbf{XLX}^T$) + $\|B\|_1$, (e) FGLR with $\mathcal{I}^X = \mathcal{I}^S = 0$, and (f) FGLR.

3) Ablation Experiment: To testify the usefulness of different modules of FGLR, including the GLR Tr($\mathbf{XLX}^T$), the sparse term $\|\mathbf{B}\|_1$, the low-rank term $R(\mathbf{S}) = \sum_{p=1}^{P}\|S\|_p$, and the Lagrangian multiplier $\mathcal{I}^X$ and $\mathcal{I}^S$, we perform the ablation experiments on Cases 2, 3, 6, and 8.

Fig. 16 displays the denoising results using different modules on the Pavia City Center data in Case 2. We can see that there is still some sparse noise and stripe noise left when only the GLR is used [see Fig. 16(c)]. When the sparse term $\|\mathbf{B}\|_1$ is added, the sparse noise is removed and only the stripe noise remains [see Fig. 16(d)]. The stripe noise is removed when the low-rank term $R(\mathbf{S})$ is introduced, shown in Fig. 16(f). In addition, the Lagrangian multipliers $\mathcal{I}^X$ and $\mathcal{I}^S$ also play significant roles in removing the noises, without which the denoising result is unsatisfactory [see Fig. 16(e)]. Table IV depicts the evaluation indexes of different modules. As we can see, each module of FGLR plays an important role in

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TABLE V
RUNNING TIME (SECONDS) OF REAL DATA EXPERIMENTS

Data	Size	LRMR	LLRSSTV	LRTDGS	RhyDe	L1HyMixDe	SSLRSSTV	DLR	HSI-SDeCNN	LRAWGTV	GSSTV	SRGLR	FGLR(Proposed)
Shang hai	$300 \times 300 \times 155$	81.1	316.6	45.2	97.4	149.7	422.5	49.5	6.0	2826.9	7948.5	1437.0	13.1
Bao qing	$300 \times 300 \times 155$	75.1	308.1	35.0	96.5	96.5	402.0	50.1	6.0	3211.4	17363.4	1153.1	11.3

realizing the removal of mixed noise, only by combining all modules correctly can we achieve satisfactory denoising results.

4) Computation Complexity and Time Consumption: Table V presents the time cost of different methods on two real datasets. Note that HSI-SDeCNN runs on the GPU, instead of the CPU. Although the deep learning-based HSI-SDeCNN costs less time than FGLR, its training process time consumption is up to 56.7 h. Besides, HSI-SDeCNN's floating-point operations (FLOPs[3]) is up to 10^{17}, and the FLOPs of FGLR is 10^9. Except for HSI-SDeCNN, FGLR thus takes the least time compared with the other methods, as shown in Table V. Combining the denoising effect and running time, we can see that our method is comprehensively superior to the other methods.

V. CONCLUSION

In this article, a fast HSI denoising and destriping method based on graph Laplacian regularization, abbreviated as FGLR, has been proposed. This method can fast and effectively remove a mixture of Gaussian noise, stripe noise, and sparse noise. Different from the traditional way of modeling the low rankness among HSI bands by the nuclear norm, we develop the graph Laplacian regularization to describe the low rankness among HSI bands. By theoretical derivation and toy experiment verification, the GLR has been proved as equivalent to the low rankness of HSI bands. Moreover, since the proposed FGLR is built on the graph with HSI bands as nodes, it spends much less time and computation resources than the state-of-the-art methods. Extensive experiments on simulated and real HSI dataset demonstrate that our method outperforms the reference methods. Future work could be focused on the extension of GLR to both the spectral and spatial domain. Besides, the proposed GLR module can be also applied to noise removal of multitemporal remote sensing data, satellite video, and so on.

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